

PV Power in the Dorm

Although I consider myself to be an environmentally concerned individual, when I entered the University of Vermont last year, I had no experience in renewable energy, or even a rudimentary knowledge of electrical wiring. And yet, one month into my first year of college life, I found myself sitting in the second row of my economics lecture with my Powerbook G4, credit card in hand, purchasing a do-it-yourself solar-energy setup from Sundancesolar.com. The decision to construct a small-scale renewable energy system had only been made about a day and a half before, when I bought wind energy credits to offset the approximate electrical usage in my suite of six freshmen students. I recalled a conversation I had had a month prior with the regional director for Smartpower, an organization that was pioneering a renewable energy purchase program for municipalities in Connecticut. While purchasing credits was one way to invest in renewable energy, he said it was infinitely preferable for municipalities to put together systems that would generate energy locally. My mind began brainstorming the possibilities. What if I created a small-scale renewable energy system at my dorm?

The Sundance setup attracted me because it contained everything I would need to activate the system including a panel (20W); a battery (sealed lead acid, 35 ampere-hours); a Morningstar SunGuard charge controller; a DC fuse; and wiring. As a novice installer, I found it was comforting to realize that all the products would work together out of the box. After ordering the system, I began planning for the setup.

One of the greatest challenges of the project was not what I was doing, but where I was doing it: in a college residence hall. I could not make any structural modifications to the building, and I was dealing with a fixed, assigned space. Nor did I have the basic power tools that I would have had access to at home. I was very fortunate in one

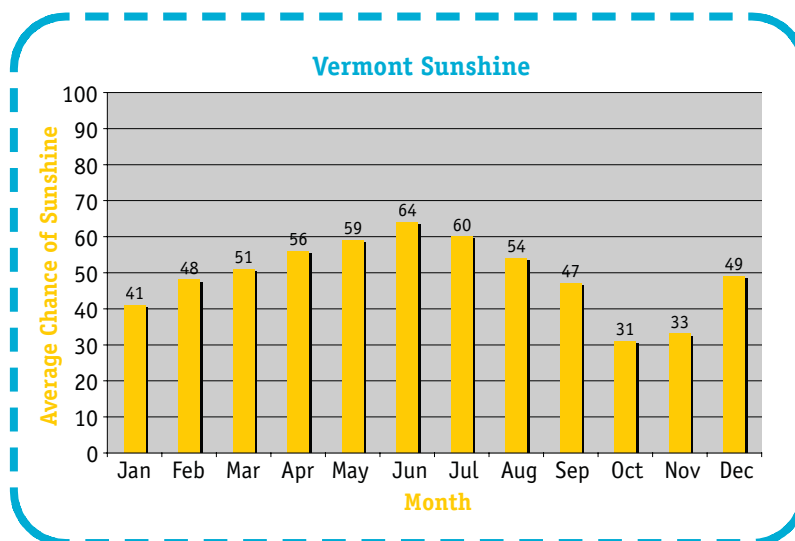
respect: I could take advantage of our suite's south-facing balcony. However, this also meant constructing a platform to hold the solar panel that would extend outward from the balcony.

After a hectic day of purchasing the plywood, screws, and wire necessary for the platform, along with the wires that would be running the energy from the balcony to my room, I was ready to begin construction. That night, I assembled the platform, which consisted of a 4 ft plywood board supported by two equally spaced 4-ft 2 x 4s. With the platform in place, I ran the wiring from the balcony to my room by extending a pole out my bedroom window and tossing the wiring from



Nizlek's 20-day challenge helped him to understand the value of conservation.

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Figure 1. Continuously sunny days are not the norm in Burlington, Vermont.

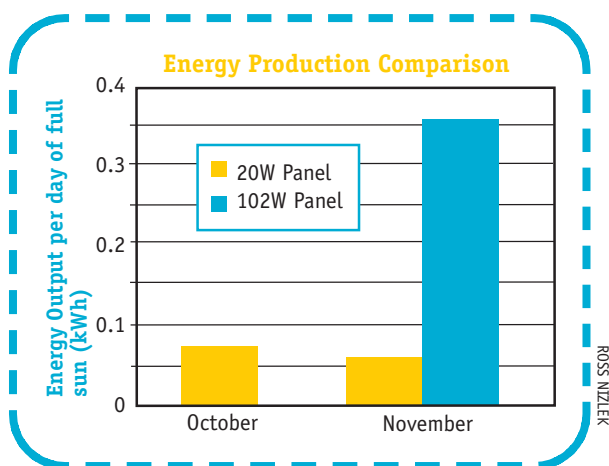


Figure 2. I purchased a larger, 102W Evergreen panel, partly because I felt the 20W panel was inadequate, and partly because I had a new idea for the project: to live off the grid for 20 days.

the balcony to the pole, which I then pulled in through the open window.

Monday morning came and the battery was delivered, but the panel and all the other components were still scanned on the truck. After a series of conversations with UPS, the driver returned with the panel at 6:30 that evening, and I began work on wiring the panel itself.

After all of these minor setbacks, it was a relief that the solar installation was uneventful. Using the instructions that Sundance provided, I connected the panel to the charge controller, and the charge controller to the battery; then I attached the voltmeter, the DC fuse, and the 12V outlet. With everything connected and the battery showing 12.5 volts, I stood back and looked at the setup, feeling empowered as I realized that I could generate my own electricity.

For the first two weeks, the system's operation during sunny weather provided me with an abundance of power, but I quickly learned that continuously sunny days are not normal weather in Burlington, Vermont (see Figure 1). When the sky clouded over, I found myself with only enough power to charge my portable electronics—including my iPod, cell phone, and PDA: running my Powerbook was out of the question most of the time.

This led me to purchase a larger, 102W Evergreen panel, partly because I felt the 20W panel was inadequate, and partly because I had a new idea for the project: to live off the grid for 20 days

(see Figure 2). I was in a room surrounded by grid electricity, but I felt that I could demonstrate how the panel offset my impact on the grid by living exclusively from it. For a 20-day period in November, I unplugged everything in my outlets, taped off the switch for the overhead light, and ran all my electrical needs through the battery, while tracking my progress on my Web site. At first I convinced myself that I had volumes of energy to spare, but I learned the necessity of conservation on the third day of the project: After working with my external monitor and all my lighting on for several hours, I checked my battery and realized that I had barely any battery left, and still needed several hours to complete my essay. I shut off all the lights, disconnected the monitor, and sat in the dark of the 12-inch screen of my laptop for the next three hours, racing to finish the paper. From that point onward, I quickly learned the art of conservation and frequently had to decide whether I'd rather watch television immediately or hedge against potentially bad weather. Many of my daily activities came to revolve around the weather forecast for the next few days.

While coming back from a day of classes to a full battery of renewable



The biggest challenge of the project was setting up the solar panel at the dorm.

energy is gratifying, the most rewarding aspect of the project has been the feedback from others. I've received a tremendous response from visitors to the project's Web site and it is heartening to know how widespread the interest in alternative energy is becoming. Despite the myriad of obstacles I was faced with, installing a solar panel in a college dormitory has been an enlightening experience, and one I would recommend without hesitation.

—Ross Nizlek

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FOR MORE INFORMATION:

To learn more about this project, please visit www.uvm.edu/%7Ernizlek/solar.